Practical 2

20% contribution to the final mark

4 questions, each has 5 points; in total $4 \times 5 = 20$ points

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Assignments

Q1 : Get the "pig.weights" data from the package "SemiPar". Fit a random intercept model and analyze the output.

Q2 : Using (3.4) derive the distribution of $\boldsymbol{Y}$ in (3.3) and show that linear mixed model (3.3) can be represented as the two level hierarchical model

$$\boldsymbol{Y}|\boldsymbol{b} \sim N_N(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{b}, \Sigma), \quad \boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D}).$$

Q3 : Show that (4.1) is the maximum likelihood estimate of $\boldsymbol{\beta}$ in the model (3.6), under the normality assumption.

Q4 : Consider the following example from the package "lme4"
Call the package "mlmRev": Examples from Multilevel Modelling Software Review
library(mlmRev)
Fit the following models and analyze the outputs in view of the differences.

$$\text{lmer(normexam} \sim 1 + (1|\text{school}), \text{data=Exam})$$

$$\text{lmer(normexam} \sim \text{standLRT} + (1|\text{school}), \text{data=Exam})$$

$$\text{lmer(normexam} \sim \text{standLRT} + (\text{standLRT}|\text{school}), \text{data=Exam})$$